

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

(192511280)

I Aluri Poojitha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Poojitha)

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20

✓

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

- ① Simple reflex agent
- ② model based agent
- ③ goal-based agent
- ④ utility based agent

Simple reflex agent

- Simple reflex agent tells about current input
- It uses the condition (if then)
- Simple reflex agent doesn't think or remember about past data.

Advantages

- It is easy to implement
- fast response

Disadvantages

- Cannot learn users habits.
- Cannot handle complex situations.

→ Model based agent

- Maintains the internal model of the environment.

→ uses the Current and previous information

Eg: Remember that bed room lights is already on.

Advantages

- Better decision making
- fast response

Disadvantages

- more memory required.

→ goal based agent → finding the ways to reach the goal

→ finding the best path to reach the goal and finding one path among multiple paths.

Advantages: we should best path to reach the goal

- intelligent planning
- flexible decision making.

disadvantages

- find the best path.
- may be more cost.
- It require more computation

Utility based agent

→ finding best path from different path, and which will consume less cost among different paths is called utility based agent.

Advantages: we can reach the destination if we know the best path.

→ Best overall decision

Eg Instead of keeping the AC at 18°C it sets it to 24°C because it saves electricity and will keep the room cool

Disadvantages and Advantages

→ Best overall decision

→ Balances Comfort, Security

Disadvantages

→ Utility Calculation is complex

→ Requires more processing power

Conclusion

It is analysed how the each type would function within the system ensures. Comfort, energy efficiency and security.

(2) 1. Uniform Cost Search (UCS)

→ finding the different ways for the solution and finding which path will be choosing according to less path.

→ uses the priority queue.

Advantages: through this we can find the minimum cost path.

→ Complete and optimal.

→ Disadvantages:

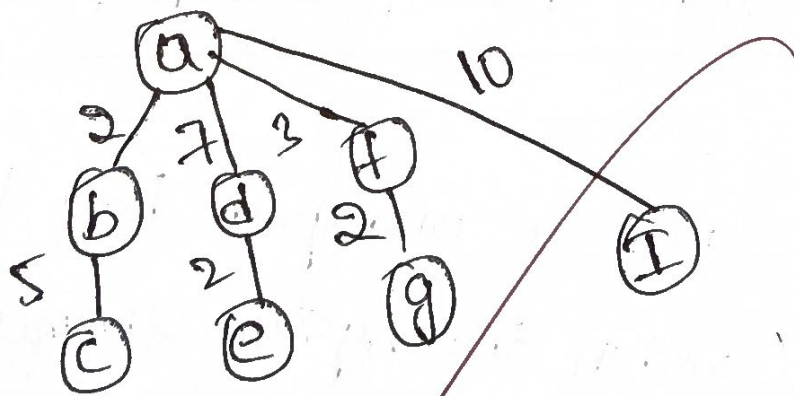
→ High memory usage

→ How big large search space

Complexity

Time:

$$O(b^l + [C^*/E])$$



a → f → g is less
because (5)

Iterative Deepening Search

→ find the problem by choosing the goal
→ fix the goal and search the problem solution
and choose the best solution.

→ Deepening search: Knowing about the solution completely and executing the solution (which knows about complete information is called fully observable).

Advantages

→ we can choose the correct path, and best.
→ fully observable.

Disadvantages

→ It consumes more time.
→ more cost.